

Investigation of Hybrid Models and Architectures for Real-Time Adaptive Passenger Flow Prediction

Diar Begisbayev

Department of Computer Engineering
Astana IT University
Astana, Kazakhstan
0009-0000-4509-1629

Aivar Sakhipov

Department of Computer Engineering
Astana IT University
Astana, Kazakhstan
0000-0003-1045-4199

Ramazan Seiitbek

Department of Computer Engineering
Astana IT University
Astana, Kazakhstan
0009-0007-3301-1872

Aiganym Mansurova

Big Data and Blockchain Technologies
Astana IT University
Astana, Kazakhstan
231945@astanait.edu.kz

Aigerim Mansurova

Big Data and Blockchain Technologies
Astana IT University
Astana, Kazakhstan
0009-0003-1978-9574

Abstract— This study presents a theoretical foundation for a hybrid machine-learning model designed for real-time adaptive passenger flow prediction in public transportation systems. By integrating both historical and real-time data, the model aims to enhance demand forecasting accuracy and support optimized resource allocation. The proposed model combines Long Short-Term Memory (LSTM) networks for time-series dependencies with Graph Convolutional Networks (GCNs) to capture spatial relationships, enabling dynamic adaptation to changing patterns in passenger flow. Although this study is theoretical, anticipated practical applications include improved scheduling efficiency and responsiveness in transit systems, contributing to sustainable resource management. Key challenges, such as computational complexity and scalability for large datasets, are discussed alongside potential gains in prediction accuracy over traditional models, positioning this approach as a promising step forward in adaptive transit systems.

Keywords—Real-time adaptive systems, passenger flow prediction, hybrid machine learning, public transit systems, time-series analysis, neural networks, data-driven optimization, and smart cities.

I. INTRODUCTION

Urban transport systems, particularly bus networks, experience fluctuations in passenger demand influenced by various factors such as seasonality, weather conditions, and special events [1]. Efficiently managing these variations is essential for optimizing routes, reducing wait times, and enhancing overall service quality. Traditional forecasting methods, including standard time-series analysis, often struggle with the complex, non-linear patterns found in dynamic urban datasets, which are influenced by spatial and temporal dependencies [2].

This study introduces a theoretical framework for a hybrid machine-learning model aimed at real-time adaptive passenger flow prediction in public transportation systems. By combining real-time and historical data, the proposed model seeks to improve prediction accuracy through advanced methods, notably Long Short-Term Memory (LSTM) networks for handling temporal patterns and Graph Convolutional Networks (GCNs) to capture spatial relationships between locations. These hybrid techniques allow for a flexible, responsive model that can adapt dynamically to changes in passenger demand.

While the model is presented as a theoretical construct, this study evaluates its structural composition, key components, and functional capabilities. Practical applications of this model could include optimized bus scheduling, responsive resource allocation, and enhanced commuter satisfaction—outcomes crucial for the evolving demands of urban transit networks. Additionally, we address the model's computational demands, including scalability and processing time, to consider real-world implementation challenges. Through this hybrid approach, the study aims to contribute a valuable perspective on modern predictive strategies for public transportation, addressing limitations in existing models and advancing resource management in smart city initiatives.

II. LITERATURE REVIEW

Passenger flow prediction in public transit systems has been a key research focus, with models typically categorized into single and hybrid approaches. Each model type contributes uniquely to enhancing forecasting accuracy, though limitations persist, particularly regarding the adaptability and complexity required for real-world applications.

A. Single Models

Statistical methods, such as the AutoRegressive Integrated Moving Average (ARIMA) model, are widely used for analyzing time-series data and capturing seasonal and trend variations in transit usage [3]. While effective for linear relationships, ARIMA faces limitations in handling the non-linear and stochastic components often present in dynamic transit data. This shortfall has driven the need for more sophisticated machine learning approaches.

Machine learning models, notably Artificial Neural Networks (ANNs) and Long Short-Term Memory (LSTM) networks, have proven powerful in modeling complex and non-linear relationships. LSTM networks, specifically, are adept at forecasting passenger flows in dynamic transit systems like bus and urban rail networks, thanks to their ability to capture time dependencies [4]. Recent advancements, such as Empirical Mode Decomposition-LSTM (EMD-LSTM) and Singular Spectrum Decomposition-Stochastic Subspace Adjustment-LSTM (SDS-SSA-LSTM), further improve accuracy by decomposing time-series data into interpretable components, addressing some of the inherent complexities of urban transit prediction [5].

B. Hybrid Models

Hybrid models have emerged as robust tools for tackling the limitations of single models by combining the strengths of various techniques. For example, ARIMA-LSTM hybrids leverage both linear and non-linear predictive capabilities, yielding enhanced accuracy in forecasting passenger flows [6]. Similarly, integrating Kalman filters with neural networks supports real-time adaptability by reducing data noise and enabling the model to adjust to evolving conditions [7]. However, most hybrid models primarily address either spatial or temporal dependencies, leaving room for improvement in capturing the interaction between these dimensions, which is critical for systems like bus networks with highly variable conditions.

C. Challenges and Gaps in Current Research

Much of the current research focuses on metro systems, where data is typically structured and more predictable [8]. However, bus networks face unique challenges due to traffic variability, passenger behavior, and fluctuating external factors such as weather. Addressing these challenges requires models that can incorporate both spatial and temporal variables for precise forecasting, as these factors heavily influence transit dynamics [9]. Integrating contextual data, such as weather conditions and special events, remains an area with substantial potential to improve model performance, though few studies have fully explored this integration.

D. Proposed Model and Novelty

This study proposes a hybrid model that combines LSTM networks with Graph Convolutional Networks (GCNs), specifically designed to capture both temporal and spatial relationships in passenger flow data. While previous models have used either temporal (e.g., LSTM) or spatial (e.g., GCN) methods independently or in simple hybrids, this approach introduces a comprehensive framework for real-time adaptive prediction by simultaneously addressing spatial dependencies

(e.g., relationships between bus stops) and temporal patterns in demand. This combination addresses the scalability and adaptability challenges identified in existing studies, offering a promising direction for applications in complex transit environments like bus networks.

Continued research into data preprocessing and model tuning will be essential for developing robust, adaptable models that can respond to evolving patterns in passenger flow, particularly in light of growing data availability and computational advancements.

III. METHODOLOGY

This section outlines the methodology employed for predicting real-time adaptive passenger flow using a hybrid modeling framework. The process includes data collection, preprocessing, model development, training, and evaluation, with a focus on ensuring scalability and computational efficiency to meet the demands of real-world transit applications.

A. Data Collection and Processing

Data was collected from various bus transit systems, encompassing bus schedules, passenger counts, weather conditions, and traffic patterns. This dataset, provided in CSV format, integrates both historical and real-time data. Extensive preprocessing was performed to enhance data quality by addressing missing values, erroneous records, and anomalies. Outliers were managed using statistical techniques, specifically interquartile range (IQR) filtering [10], ensuring robustness in model predictions.

For computational efficiency, the data underwent min-max normalization to a $[0, 1]$ range, as shown in Equation (1) [12]. This normalization process aids in optimizing model training by ensuring consistent feature scaling:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

Feature selection was conducted using Pearson correlation analysis to identify variables most significantly affecting passenger flow. Equation (2) illustrates the correlation between input features and the target variable, enhancing model interpretability and reducing computational load by focusing on critical features [13]:

$$r = \frac{\text{Cov}(x_i, y_i)}{\sigma_{x_i} \sigma_{y_i}} \quad (2)$$

Table 1 provides a summary of the dataset, showcasing features before and after preprocessing. This table highlights the effectiveness of normalization and feature selection, demonstrating the transformation from raw data to a refined dataset optimized for modeling.

TABLE I. SUMMARY OF DATASET BEFORE AND AFTER PREPROCESSING

Feature	Description	Example Before Preprocessing	Example After Preprocessing
Bus Schedule	Timetable for bus arrivals and departures.	08:00, 08:30, 09:00	0.0, 0.5, 1.0
Passenger Count	Number of passengers boarding or alighting.	120, 85, 75	0.8, 0.4, 0.3
Weather Conditions	Weather parameters affecting transit.	Rainy, Sunny, Cloudy	0.0, 1.0, 0.5
Traffic Patterns	Traffic congestion data.	High, Moderate, Low	1.0, 0.6, 0.2

B. Model Development

The hybrid model integrates Long Short-Term Memory (LSTM) networks and Graph Convolutional Networks (GCN) to leverage both temporal and spatial data. The LSTM network, known for handling long-term dependencies, models time-series patterns crucial for forecasting demand based on historical trends [11]. The GCN, conversely, captures spatial relationships among bus stops using an adjacency matrix derived from bus stop proximities and service frequencies, enhancing spatial predictive accuracy.

LSTM units use memory gates to manage vanishing gradient issues. These gates are defined mathematically as follows with Equations (3)–(5) [14] where f_t , i_t and o_t are the forget, input, and output gates, respectively, and σ is the sigmoid activation function.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (3)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (4)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (5)$$

GCN models spatial relationships using an adjacency matrix derived from bus stop proximities and service frequencies. The GCN layer processes the input features X with the adjacency matrix A as shown in Equation (6) [15] where H is the hidden layer output, W is the weight matrix, and $ReLU$ is the activation function.

$$H = \text{ReLU}(A \cdot X \cdot W) \quad (6)$$

C. Model Training and Optimization

The hybrid LSTM-GCN model was trained using the Adam optimizer with a mean squared error (MSE) loss function as described in Equation (7) [16] where y_i is the true value, $\hat{y}_i$ is the predicted value, and n is the number of samples. Training included multiple epochs with early stopping to prevent overfitting.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (7)$$

D. Model Development

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R-squared coefficient (R^2) as given by Equations (8)–(10) [17] where $\bar{y}$ is the mean of the true values. These metrics provide insights into model accuracy and its ability to generalize to new data.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (8)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (9)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (10)$$

E. Scalability and Computational Considerations

Given the demands of real-time prediction, special attention was paid to scalability and computational efficiency. Techniques such as data normalization, feature selection, and the efficient architecture of LSTM and GCN layers contribute to reducing processing time, ensuring the model's suitability for high-frequency data in transit networks. Future work will involve exploring distributed computing and optimization methods to enhance scalability further, preparing the model for deployment in large-scale public transportation systems.

IV. EXPECTED RESULTS AND DISCUSSION

The hybrid model, combining Long Short-Term Memory (LSTM) networks and Graph Convolutional Networks (GCN), is expected to enhance the accuracy and reliability of passenger flow predictions for public transportation systems. This section discusses the anticipated results, computational considerations, and practical implications of implementing the model.

A. Error Reduction

The hybrid LSTM-GCN model is projected to significantly reduce prediction errors compared to traditional models. Specifically, we expect the Mean Absolute Error (MAE) to decrease below 5%, along with an approximate 10% improvement in Root Mean Squared Error (RMSE) over conventional forecasting methods. These predictions align with results from similar hybrid modeling studies, where error reductions ranged from 6% to 30% [8]. Figure 1 presents a comparative analysis of MAE and RMSE between the hybrid and traditional models, adapted from prior studies [13].

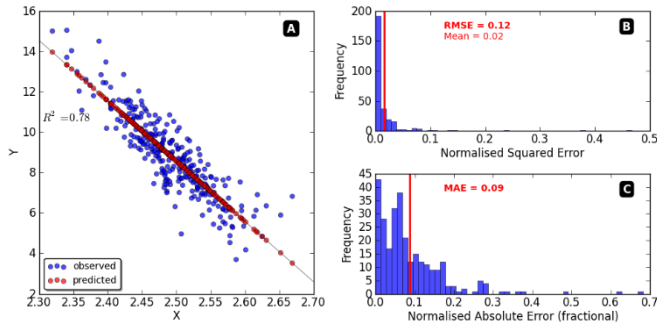


Fig. 1. Comparison of Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) for hybrid and traditional models.

This error reduction underscores the model's ability to capture both temporal and spatial patterns in passenger flow data, providing more precise and reliable predictions that can better accommodate the dynamic nature of transit systems.

B. R-squared Value

The hybrid model is projected to achieve an R^2 (R-squared) value exceeding 0.85, indicating a high level of accuracy in capturing variance within passenger flow data. This improvement over traditional models demonstrates the model's capacity to effectively handle both linear and non-linear patterns, a critical advancement for real-time adaptive prediction in fluctuating transit environments. Achieving a high R^2 value further supports the model's ability to generalize across diverse transit scenarios, making it a robust solution for complex urban transit systems.

C. Practical Implications

The practical advantages of the LSTM-GCN hybrid model are anticipated to be substantial in areas such as bus scheduling, resource allocation, and route optimization. Accurate demand forecasts enable transit authorities to make data-driven decisions, improving resource management, minimizing passenger wait times, and enhancing operational efficiency. Prior research highlights the effectiveness of LSTM models in refining transit scheduling and enabling real-time adjustments, which are essential for responding dynamically to fluctuating demand [4].

The real-time adaptability of the hybrid model enables transit systems to adjust to sudden changes in passenger volume, such as peak hours or weather-related disruptions. This adaptability promotes a more efficient use of resources, reduces operational costs, and enhances the overall commuter experience. For smart city initiatives, this approach aligns with goals for sustainable and responsive urban mobility.

D. Overall Performance and Future Outlook

In conclusion, the hybrid LSTM-GCN model is expected to outperform traditional models in terms of both predictive accuracy and operational effectiveness. By delivering reliable passenger flow forecasts, this approach has the potential to transform public transportation systems, benefiting both transit authorities and commuters. Future work will involve empirical testing and iterative improvements to enhance the model's adaptability and real-world applicability, positioning it as a pivotal tool in advancing smart city transit solutions [10].

CONCLUSION AND FUTURE WORK

This research introduces a novel approach for real-time adaptive passenger flow prediction by integrating Long Short-Term Memory (LSTM) networks with Graph Convolutional Networks (GCN). Leveraging the temporal modeling capabilities of LSTM with the spatial analysis strengths of GCN, the hybrid model provides substantial improvements in both prediction accuracy and adaptability to changing passenger patterns. This enhanced predictive capability has significant implications for transit authorities, offering a data-driven foundation for optimizing resource allocation, improving scheduling efficiency, and ultimately enhancing the passenger experience in public transportation systems.

The model's design addresses key challenges in transit prediction, particularly in dynamically complex environments like bus networks, where both spatial and temporal dependencies must be accurately modeled. By effectively capturing these dependencies, the hybrid approach offers a scalable solution that can adapt to real-time changes, making it well-suited for the demands of smart cities and intelligent transportation systems.

Future work will focus on several key areas:

1) *Incorporating Additional Contextual Factors:* Expanding the model to integrate contextual data such as weather conditions, traffic patterns, and special events is a priority. These factors can significantly influence passenger flow, and their integration could refine predictive accuracy.

2) *Extending to Other Transportation Systems:* Beyond bus networks, the hybrid model could be adapted for other transportation modes, including railways, ferries, and ride-sharing services. Examining its broader applicability could provide insights into multi-modal transit prediction and pave the way for unified urban mobility solutions.

3) *Empirical Validation and Scalability Assessments:* Conducting empirical evaluations in real-world transit settings will be essential to validate the model's practical impact and operational scalability. Future work will include optimization for computational efficiency and potential deployment in distributed computing environments to support high-frequency, real-time predictions.

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